The Van der Pol Oscillator: Comparing Numerical and Machine Learning Approaches

Daniel Shamsoddini

Equation

The Van Der Pol oscillator is a nonconservative oscillator with nonlinear damping. It is a second order ODE.

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$$

The forced Van Der Pol oscillator is non-autonomous

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x - A\sin(\omega t) = 0$$

Importance

The Van Der Pol oscillator and its variants commonly show up in

- Electrical circuits
- Biological Oscillating Systems (Neurons, Fitzhugh Nagumo model)
- Nonlinear Systems

It lacks an analytical solution, especially the Forced one

- We have to use methods to approximate it

Numerical Methods

Written as a system of first order ODEs, different numerical methods can be used to approximate the solution to the Van Der Pol oscillator differential equations

$$\dot{x} = y$$
$$\dot{y} = \mu(1 - x^2)y - x$$

Euler Method

- First order method, very fast and straightforward, less accurate and less stable

RK4

- Fourth order method, slightly slower, generally more stable

ML Approaches

However, if we do not know the coefficients or underlying equations, or we wish to not have to evaluate a large number of timesteps to find a trajectory value at a point in time

Numerical methods fall short in this regard

I will be investigating the utility and accuracy of three ML based methods:

- Feedforward Neural Network
- Neural ODE
- Physics Informed Neural Network

Traditional ANN

We simply train a neural network to predict the trajectory at a given time

$$t \rightarrow \mathrm{NN} \rightarrow (x(t), \dot{x}(t))$$

Advantages:

- Computationally cheap
- Easy to train
- Only requires trajectory data points if training with regression

Disadvantages:

- No understanding of physics
- May diverge outside of its training horizon

Neural ODE

A Neural ODE is a continuous model that attempts to learn the vector field that represents a differential equation

$$\hat{x} = f_\theta(x)$$

Advantages:

- Stable over long time trajectories
- Learns the actual differential equation

Disadvantages:

- Requires an ODE solver during training
- Still no Physics based constraints or physically meaningful constraints
- Very computationally expensive

PINN

Physics Informed Neural Networks (PINNs) attempt to train a traditional neural network while using the differential equation or physics principles as a part of loss calculation

$$t \rightarrow \mathrm{NN} \rightarrow (x(t), \dot{x}(t))$$

$$Loss = Loss_{data} + Loss_{DE}$$

Advantages

- Requires less or even no trajectory data points
- Enforces consistency in the NN results
- Produces results tied to physics/DE properties

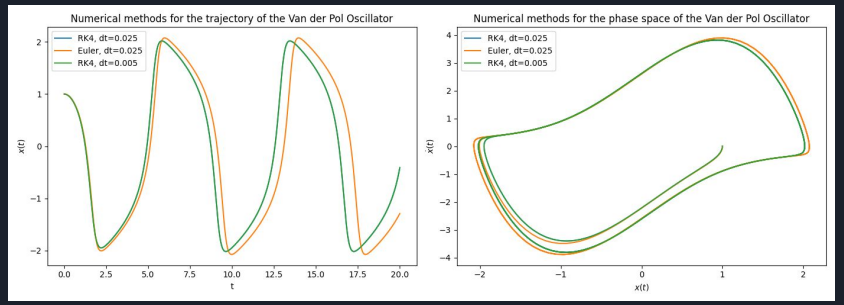
Disadvantages

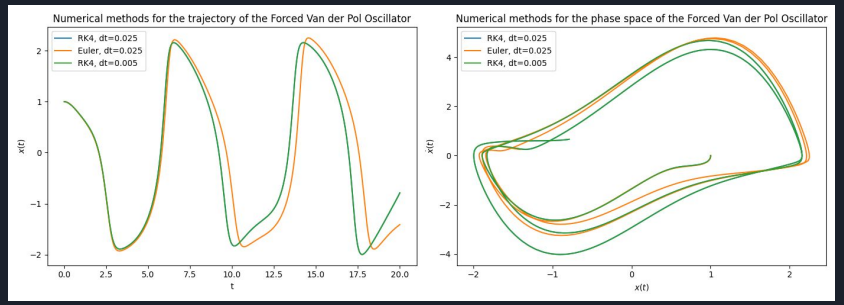
- Requires the underlying differential equation
- Training can be slow and sensitive to hyperparameters

Results

Numerical Methods:

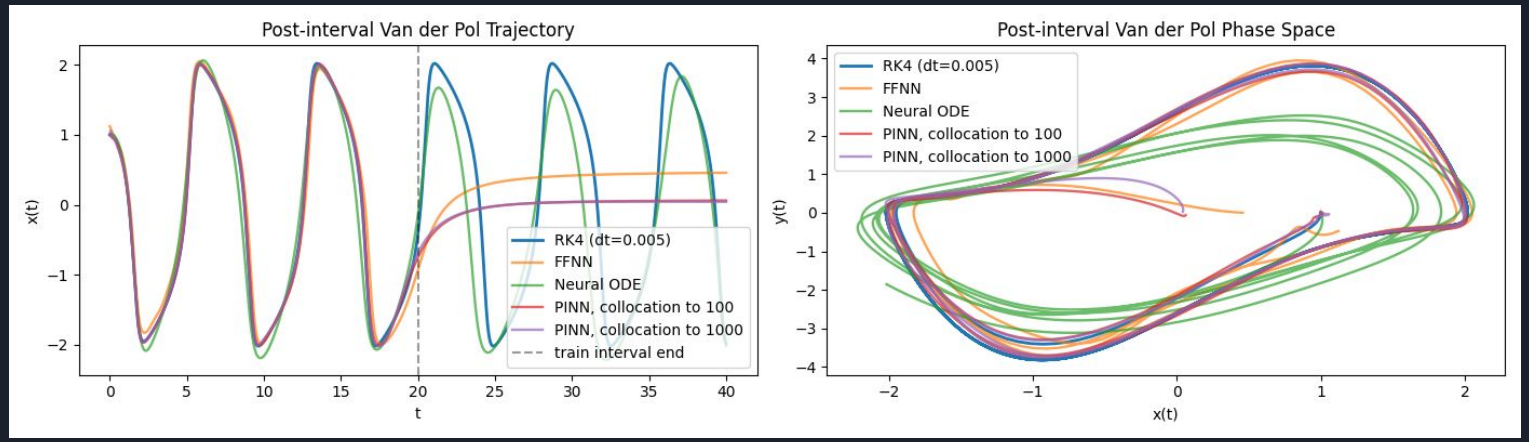
- We can see that the explicit Euler method error is noticeable compared to the RK4 for the forced VDP oscillator in both trajectory and phase space
- For the unforced VDP, holding all parameters equal, this difference is less noticeable, but still present





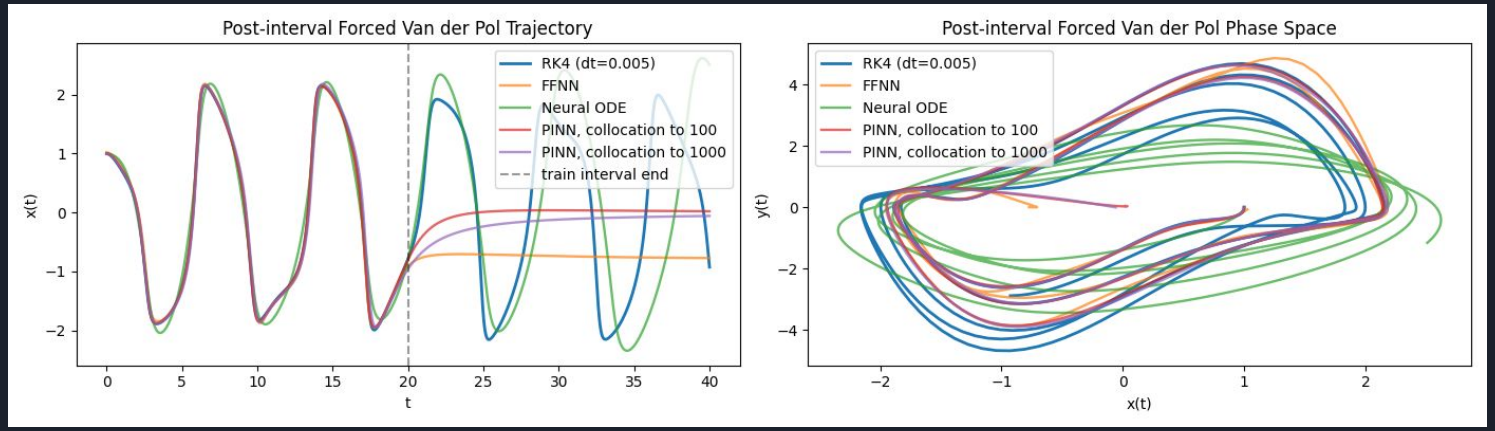
Results

As such, using noisy data collected from the RK4 method, networks were trained, resulting in



Results

For the forced VDP, the results are similar but worse due to the non-autonomous forcing introducing additional complexity



Conclusion

While machine learning approaches are useful for modeling ODE systems, and can succeed on known intervals, they struggle with extrapolation past those time intervals

- Numerical methods remain much more reliable for accurate trajectory prediction

Generally, Neural ODEs perform the best of all the Machine Learning methods at extrapolation, but if you know your interval PINNs are a good bet

- Neural ODEs benefit from short time-steps and possibly a more accurate numerical method?

Both methods have their uses and the Van Der Pol oscillator is far from an unmodelable system, but caution must be exercised in model assumptions and extrapolation attempts